

Galilean Transformation and Lorentz Transformation Equations



MD. GULAM MUSTAFA
Lecturer (Physics)

Department of Science & Humanities

Bangladesh Army International University of Science and Technology (BAIUST)
Syedpur, Cumilla-3501

Whatever we will discuss

- Galilean Transformations and its explanation.
- Limitations of Galilean Transformations.
- Equations of Lorentz Transformations and Inverse Lorentz Transformations.
- Speciality of Lorentz Transformations

Galilean Transformations

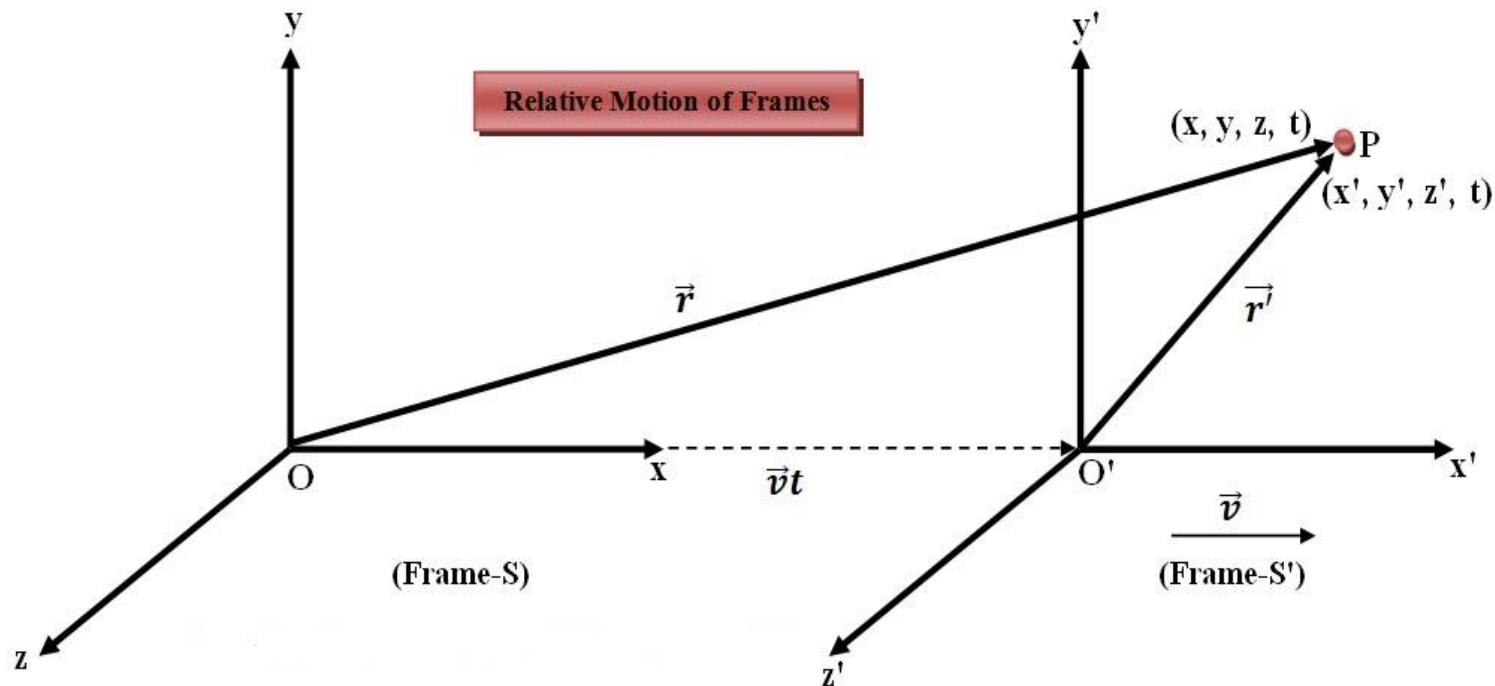
If an event is observed by two observers from two separate inertial frames, then naturally two sets of coordinates will be obtained for the observers of two different frames. The equation which is obtained in order to establish the relation between the two sets of coordinates for that event is called the Galilean transformation equation or Galilean Transformation.

$$x' = x - vt$$

$$y' = y$$

$$z' = z$$

$$t' = t$$



Limitations of Galilean Transformations

Limitation-1: According to the first postulate of special relativity, “*The laws of physics have the same form in all inertial frames of reference*”. But, in case of electricity and magnetism, the equations have different forms for different frames of reference after Galilean transformation.

Limitation-2: According to the second postulate of special relativity, “*The speed of light in free space has the same value in all inertial frames of reference*”. But, it is shown that the speed of light gives different values for different frames of reference after Galilean transformation.

Lorentz Transformation Equations

Lorentz Transformation Equations

$$x' = \frac{x - vt}{\sqrt{1 - v^2/c^2}}$$

$$y' = y$$

$$z' = z$$

$$\text{and } t' = \frac{t - vx/c^2}{\sqrt{1 - v^2/c^2}}$$

Inverse Lorentz Transformation Equations

$$x = \frac{x' + vt'}{\sqrt{1 - v^2/c^2}}$$

$$y = y'$$

$$z = z'$$

$$t = \frac{t' + vx'/c^2}{\sqrt{1 - v^2/c^2}}$$

Speciality of Lorentz Transformations

Speciality-1: According to the Galilean transformations, Time in all frames of reference is the same (i.e. $t' = t$). But Lorentz transformations show that time is not same for all frames of reference. i.e., It gives different values for different observers of different reference frames.

$$i.e. \quad t' = \frac{t - vx/c^2}{\sqrt{1 - v^2/c^2}}$$

Speciality-2: If $v \ll c$; then Lorentz transformations approach to Galilean transformations. Thus we may say that Lorentz transformations hold for all velocities; while Galilean transformations hold only for low velocities. Consequently Lorentz transformations are superior to Galilean transformations.



Thank You
Thanks for your attention